7.5 Factoring $x^2 + bx + c$

Name _____ Period ____ Date _____

Algebra I 7.5 · Student sheet

I can…

- factor $x^2 + bx + c$ when c is positive.
- factor $x^2 + bx + c$ when c is negative.
- solve equations and real-life problems by factoring.

Vocabulary · fill in the missing word

Factoring $x^2 + bx + c$ — _____ it as $(x + p)(x + q)$, where $p + q = b$ and $pq = c$

Sign rules — $c > 0$: p and q have the same sign as b. $c < 0$: p and q have _____ signs; the larger one has the sign of b

Part 1 · Product–sum puzzles

Find two numbers with the given product and sum.

Product	Sum	The two numbers
12	7	
12	−7	
−12	1	
−12	−4	
18	9	
−20	1	

Multiply: $(x + 3)(x + 4)$ = _____

Where do the 7 and the 12 come from? _____

Part 2 · Notes: fill in the blanks

$x^2 + bx + c = (x + p)(x + q)$, where $p \cdot q$ = _____ and $p + q$ = _____.

When c is positive, p and q have the _____ sign, the sign of _____.

When c is negative, p and q have _____ signs; the one with the larger absolute value has the sign of _____.

Always check by _____.

Part 3 · Examples (work them with the class)

Example 1

Factor $x^2 + 7x + 10$.

Example 2

Factor $x^2 - 9x + 20$.

Example 3

Factor $x^2 + 3x - 28$.

Example 4

Factor $x^2 - 2x - 24$.

Example 5

Solve $x^2 - 4x = 21$.

Example 6

A rectangular patio has an area of $x^2 + 11x + 24$ square feet. One side is $(x + 8)$ feet. Write a binomial for the other side.

Part 4 · Partner practice

Partner 1

Factor $x^2 - 11x + 18$.

Partner 2

Solve $x^2 + 5x - 14 = 0$.